

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1338

Roxbury 138 kV
March 06, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1338. The Interconnection Customer proposes a 244.3 MW solar photovoltaic facility interconnecting at the Roxbury 138 kV substation in the Pennsylvania area. The request was received on November 17, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 463 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$39,019,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
REWEY to LOCO HILLS 138 kV	MILWAUKEE to REWEY 138 kV	164.4	180	18.4
MILWAUKEE to REWEY 138 kV L	REWEY to LOCO HILLS 138 kV	144.6	307	25.3
BUXTON to NORTHSIDE DRIVE 138 kV	FT HUMBUG 345-138 kV Auto	107.6	149	6.0
PINOLA 09 to ROXBURY 138 kV	ROXBURY to SOUTH BAY 1 138 kV	106.8	222	100.0
ROXBURY to SOUTH BAY 1 138 kV	ROXBURY to PINOLA 09 138 kV	105.2	226	100.0
YEAGERTOWN to LEWISTOWN 138 kV L	WEST CAMPUS to EDES TAP 2 345 kV L	105.1	207	6.5
TAP to PINOLA 09 138 kV	ROXBURY to S BAY 1 138 kV	104.8	210	100.0
TAP to BOXWOOD 138 kV line	S BAY 1 to ROXBURY 138 kV	103.5	204	100.0
FAYETTEVILLE to GRAND POINT 138 kV	ROXBURY to S BAY 1 138 kV Line	103.0	173	100.0
BOXWOOD SUBSTATION to DPC ARGYLE 138 kV	SOUTH BAY 1 to ROXBURY 138 kV line	101.4	199	100.0
REWEY to S BAY 1 138 kV Line	PINOLA 09 to ROXBURY SUBSTATION 138 kV	100.4	232	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
REWEY to LOCO HILLS 138 kV	Rebuild line with higher-capacity conductor	\$14,568,000	100% (\$14,568,000)
MILWAUKEE to REWEY 138 kV L	Rebuild line with higher-capacity conductor	\$5,073,000	72% (\$3,630,000)
BUXTON to NORTHSIDE DRIVE 138 kV	Reconductor line	\$2,827,000	72% (\$2,023,000)
PINOLA 09 to ROXBURY 138 kV	Reconductor line	\$4,898,000	100% (\$4,898,000)
ROXBURY to SOUTH BAY 1 138 kV	Reconductor line	\$3,683,000	100% (\$3,683,000)
YEAGERTOWN to LEWISTOWN 138 kV L	Reconductor line	\$2,167,000	73% (\$1,587,000)
TAP to PINOLA 09 138 kV	Reconductor line	\$3,747,000	100% (\$3,747,000)
BOXWOOD to TAP 138 kV line	Reconductor line	\$1,227,000	100% (\$1,227,000)
GRAND PT to FAYETTEVILLE 138 kV	Reconductor line	\$881,000	100% (\$881,000)
BOXWOOD SUBSTATION to DPC ARGYLE 138 kV	Reconductor line	\$667,000	100% (\$667,000)
REWEY to S BAY 1 138 kV	Reconductor line	\$2,108,000	100% (\$2,108,000)
	Total allocated to project		\$39,019,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.